

Module 3

VPC in GCP



Goals: Understand about VPC, IP address, subnet, firewall

- VPC overview
- Global network and subnets
- IP address
- Firewall rules

VPC overview:

- A [Virtual Private Cloud \(VPC\)](#) network is a virtual version of a physical network, such as a data centre network.
- VPC provides connectivity for your [Compute Engine virtual machine \(VM\) instances](#), [Google Kubernetes Engine \(GKE\) clusters](#), [App Engine flexible environment instances](#), and other resources in your [project](#).
- New projects start with a default network that has one subnetwork (subnet) in each region.
- Projects can contain multiple VPC networks.

VPC networks have the following properties:

- ▶ VPC networks, including their associated routes and firewall rules, are **global resources**. They are *not* associated with any particular region or zone.
- ▶ Subnets are **regional resources**. Each subnet defines a range of IP addresses.
- ▶ Traffic on instances can be controlled with network **firewall rules**.
- ▶ Network administration can be secured by using **Cloud Identity and Access Management (Cloud IAM)** roles.
- ▶ An **organization** can use **Shared VPC** to keep a VPC network in a common host project.
- ▶ VPC networks can be securely connected in hybrid environments by using **Cloud VPN** or **Cloud Interconnect**.

- Each VPC network consists of one or more useful IP range partitions called subnets.
- Each subnet is associated with a region. VPC networks do not have any IP address ranges associated with them. IP ranges are **defined for the subnets**.
- A network must have at least one subnet before you can use it.
- Auto mode VPC networks create subnets in each region automatically.
- Custom mode VPC networks start with no subnets, giving you full control over subnet creation.
- User can create more than one subnet per region.

Shared VPC

Scenario: Your organization has multiple projects. You want resources in different projects to talk to each other?

How to allow resources in different projects to talk with internal IPs securely and efficiently?

Enter **Shared VPC**

Created at organization or shared folder level (Access Needed: Shared VPC Admin)

Allows VPC network to be shared between projects in same organization
Shared VPC contains one host project and multiple service projects:

Host Project - Contains shared VPC network

Service Projects - Attached to host projects

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VPC Peering

Subnet overview:

- ▶ Google Resources on Same VPC & Different Subnets, Resources can access each other over internal IP addresses.
- ▶ Each VPC network has its own distributed **firewall**, which allows or denies traffic into and out of the network, and between VM instances.
- ▶ Within Network, resources can access reach each other using Internal IP, because they are from trusted party.
- ▶ Resources of Other Network treated as External Resources and can be access only external IP only, even they are part of same project.

Subnet overview:

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- ▶ Resources of Other Network treated as External Resources and can be access only external IP only, even they are part of same project.
 - Subnet IPs cannot be overlapped.
 - us-east1 - 10.230.8.0/24
 - ~~us-west1 - 10.230.8.0/12~~

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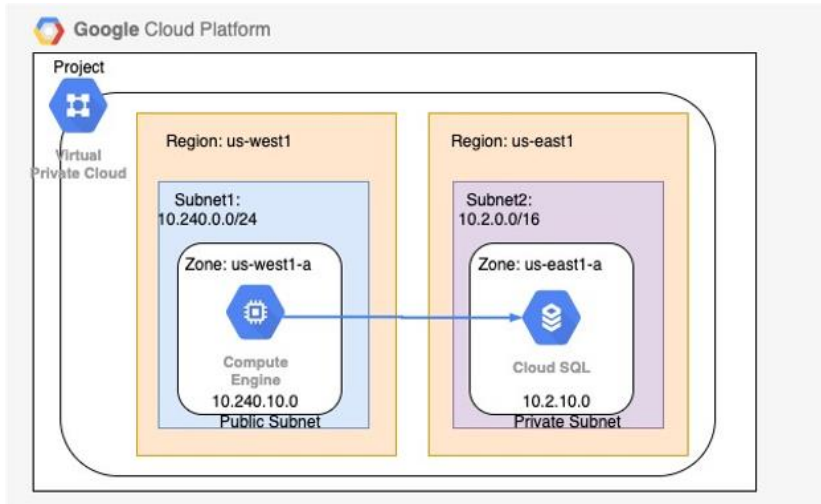
How do you **separate public resources from private resources** inside a VPC?

(Solution) **Create different subnets** for public and private resources

Resources in a public subnet **CAN** be accessed from internet

Resources in a private subnet **CANNOT** be accessed from internet

BUT resources in public subnet can talk to resources in private subnet



How subnet connect to outside internet

- Cloud NAT: Cloud NAT is a network address translation (NAT) service that allows VMs in private subnets to connect to the internet. Cloud NAT creates a public IP address for each VM in the private subnet, and then routes traffic from the VM to the internet through the public IP address.
- External IP addresses: VMs in public subnets can be assigned external IP addresses. External IP addresses are public IP addresses that are assigned to VMs in public subnets. This allows VMs in public subnets to connect directly to the internet without using Cloud NAT.
- VPC Network Peering: VPC Network Peering allows you to connect two VPC networks together. This allows VMs in one VPC network to connect to VMs in the other VPC network, even if they are in different subnets.
- Cloud VPN: Cloud VPN is a VPN service that allows you to connect your on-premises network to your GCP network. This allows VMs in your on-premises network to connect to VMs in your GCP network, even if they are in different subnets.

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VPC Peering

Scenario: How to connect VPC networks across different organizations?

Enter VPC Peering

Networks in same project, different projects and across projects in different organizations can be peered

All communication happens using internal IP addresses

Highly efficient because all communication happens inside Google network
Highly secure because not accessible from Internet

No data transfer charges for data transfer between services

IP address overview

- IP address are assigned to Compute resources.
- Each VM must have an associated IP Address. An internal IP address and an external IP address.
- The internal IP address is used to communicate between instances in the **same VPC network**.
- The external IP address is used to communicate with instances in other networks or the Internet.

Firewall

- ▶ Firewall rules let you **allow or deny traffic** to and from your virtual machine (VM) instances based on a configuration that you specify.
- ▶ Firewall protects your VM from **unapproved traffic & connection**.
- ▶ Enabled Google Cloud firewall rules are always enforced, protecting your instances regardless of their configuration and operating system, even if they have not started up.
- ▶ Firewall rules are defined at the network level, connections are allowed or denied on a per-instance basis.

► Pre-populated rules in the default network

► [default-allow-internal](#) - Allows ingress connections for all protocols and ports among instances in the network. This rule has the second to-lowest priority of 65534, and it effectively permits incoming connections to VM instances from others in the same network.

► [default-allow-ssh](#)

Allows ingress connections on TCP port 22 from any source to any instance in the network. This rule has a priority of 65534.

► [default-allow-rdp](#)

Allows ingress connections on TCP port 3389 from any source to any instance in the network. It enables connections to instances running the Microsoft Remote Desktop Protocol (RDP).

► [default-allow-icmp](#)

Allows ingress ICMP traffic from any source to any instance in the network. It enables tools such as ping.

Ingress

- Ingress firewall rules control incoming connections from a source to target instances in your VPC network. The source for an ingress rule can be defined as-
 - A range of IPv4 addresses.
 - Other instances in your VPC network identified by service account
 - Other instances in your VPC network identified by network tags

Egress

- Egress firewall rules control outgoing connections from target instances in your VPC network.
- User can permit outbound traffic to specific destinations, such as a range of IPv4 addresses, on protocols and ports that you specify.
- Every egress rule needs a *destination*. The default destination is any IP address (0.0.0.0/0), but you can create a more specific destination by using a range of IPv4 addresses in CIDR format.

Thank you

